

Project Proposal

Hullabaloo Party Supplies (HPS) Customer Order System

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Project Overview

Hullabaloo Party Supplies (HPS) is a nationwide party supply provider and distributor operating across thirteen locations in the United States. Historically, HPS has operated as a business-to-business supplier, selling party goods exclusively to third-party resale vendors. However, in an effort to enhance profitability and strengthen brand presence, HPS is transitioning to a direct-to-consumer (D2C) sales model. This shift to a D2C approach offers the organization several benefits (Spyker, 2022):

- Shorter time-to-market
- Higher profit margins
- Owning customer experience
- Better control of pricing
- Additional stock management control
- Enforce a high level of quality standards
- Faster and more reliable scaling
- Enhance consumer data

To support this new model, HPS requires the development of a robust customer ordering system. This system will serve as the backbone of the online shopping experience, allowing consumers to browse products, manage shopping carts, create accounts, select shipping options, and complete payments securely. The development of this system is critical to the successful execution of HPS's D2C strategy, aiming to enhance consumer engagement and drive business growth.

Problem Statement

The existing order processing system at HPS was developed specifically to accommodate orders from resale vendors. As a result, it lacks the necessary features to support individual consumer transactions, including a shopping cart function, customer account creation, tax and shipping calculations, and integrated payment processing. Because the current system cannot be adapted to meet these requirements, the organization must develop a new customer ordering system. Without this system, HPS will not be able to execute its D2C strategy resulting in potential revenue loss, customer dissatisfaction, and a competitive disadvantage.

Goals and Objectives

To ensure the new customer ordering system supports organizational needs, aligns with the corporate strategic plan, and reinforces the company's competitive strategy, HPS must first establish a clear list of goals and objectives (Valacich & George, 2020).

Goal 1: Create a Seamless Online Ordering Experience for Consumers

A smooth, intuitive online shopping experience is essential for attracting and converting individual consumers. By making it easy to browse, select, and purchase products, HPS can reduce friction, maximize customer satisfaction, and increase sales completion rates.

- **Objective 1.1:** Design and deploy a user-friendly website interface that enables customers to browse, select, and add multiple items to a shopping cart within three clicks. Although the number of clicks does not directly correlate with user satisfaction or sales success, the Three-Click Rule helps designers focus on the information users need and could help them create a better, more streamlined website (Porter, 2003).
- **Objective 1.2:** Streamline checkout process aimed at achieving a cart completion rate of 50% within the first 90 days of launch, with a goal of reaching 70% checkout completion after

optimization. According to a study performed by Bold Commerce (2021), top-performing brands achieved a desktop checkout completion rate of 69.25% while average performers had a rate of 47.45%.

Goal 2: Support Long-Term Customer Engagement and Retention

Sustained customer relationships are crucial for ongoing success in a D2C model. Features that encourage account creation and personalization of the post-purchase experience can improve retention and long-term value.

- **Objective 2.1:** Enable optional customer account creation, with features for order history and saved shipping addresses that are available at launch. “Online Service Personalization: Impact on Customer Satisfaction and Loyalty in E- Commerce Platforms” (2024) highlights how personalized experiences can lead to increased customer engagement and retention.
- **Objective 2.2:** Develop an automated email system for order confirmation, shipping updates, and personalized marketing within 60 days of system deployment. These personalized services can significantly enhance customer loyalty, emphasizing the importance of implementing personalized email campaigns as part of an automated system to foster stronger customer relationships (Tumpa, 2024).

Goal 3: Ensure Secure and Flexible Payment Processing

Offering multiple payments options builds trust and supports purchasing convenience. HPS can minimize transactional errors and have confidence in maintaining standards through compliance testing.

- **Objective 3.1:** Integrate secure payment gateways (credit cards & PayPal) in compliance with PCI DSS standards before the system goes live. HPS should outsource payment gateways to third-party providers who are already in PCI DSS standards (Best Practices for Securing E-commerce Special Interest Group & PCI Security Standards Council, 2017).
- **Objective 3.2:** Conduct at least 2 rounds of payment transaction testing to ensure 100% accuracy in tax, total cost calculations, and provide a seamless user experience. The assessments should cover payment testing to evaluate user experience and payment gateway testing to ensure the technical integration of gateway systems (Testlio, 2025).

Goal 4: Establish Accurate and Real-Time Inventory Management

Real-time inventory tracking ensures that consumers are only able to order available products, preventing operational issues. Integrating with the inventory system will improve accuracy, reduce manual effort, and streamline fulfillment.

- **Objective 4.1:** Connect the ordering system to HPS’s inventory database to reflect live products availability in less than a 5-minute delay. HPS can avoid overselling products resulting in reducing stockouts, maintaining credibility, and preserving customer trust (Speed Commerce, 2024).
- **Objective 4.2:** Implement automated inventory reconciliations across sales channels and warehouses to ensure accuracy and prevent discrepancies. This improves demand forecasting, reduces fulfillment errors, and supports operational efficiencies (Team Omniful, 2024).

These goals and objectives will guide the design and development of a customer ordering system that meets operational requirements and strategic priorities.

Key Assumptions

Though certainty cannot be fully confirmed, key assumptions must be made to prevent the derailing of HPS's customer ordering system implementation. These assumptions are factors that are believed to be true or certain throughout the development lifecycle. Key assumptions can benefit HPS as they will enable them to effectively manage uncertainties, identify early warning signs that necessitate quick action, and foster common understanding of project foundations which can reveal unknown risk, constraints, and new premises (Smartsheet, n.d.).

The following are key assumptions associated with the HPS development of the customer ordering system:

- The system will be developed internally by HPS's Systems Analyst Team, rather than a third-party vendor.
- All necessary project team members will be available as planned throughout the project phases.
- The existing technology infrastructure is compatible with the new customer order system and will support integration.
- Risks and potential roadblocks are adequately identified and addressed.
- Budget allocated for the project will be sufficient and approved to cover costs of development, testing, deployment, and training.
- Project requirements gathered during analysis will remain relatively consistent with no major changes or scope creep.
- There will be no major browser updates that will significantly affect website display or functionality (Valacich & George, 2020; Smartsheet, n.d.).

These assumptions provide a foundation for realistic planning and risk management. While not guaranteed, acknowledging them helps align expectations and prepares HPS for potential adjustments during progression.

Stakeholders and Responsibilities

It is essential for HPS to identify all key stakeholders and clearly define their roles and responsibilities. This will ensure that each party understands their part in meeting project objectives and aligning the solution to organizational needs (Valacich & George, 2020). Table 1 is a breakdown of HPS's stakeholders involved and their related responsibilities.

Table 1

Key Project Stakeholders and Their Responsibilities

Key Stakeholders	Responsibilities
Executive Leadership	• Approves project funding and strategic direction • Resolves high-level issues and ensures alignment with corporate goals.
System Analyst Team	• Lead system design, development, and testing • Coordinate integration with inventory and payment systems
Finance and Accounting (F&A)	• Validates payment processing accuracy and tax calculations • Ensures financial reporting compliance

IT Support	- Assists with infrastructure setup, cybersecurity, and hosting - Ensures ongoing maintenance and uptime
Marketing and Sales	- Provides input on user experience and customer engagement - Uses system data for personalized marketing efforts
Customer Service	- Offers insights on customer system complaints - Participates in user acceptance testing (UAT) and provides feedback

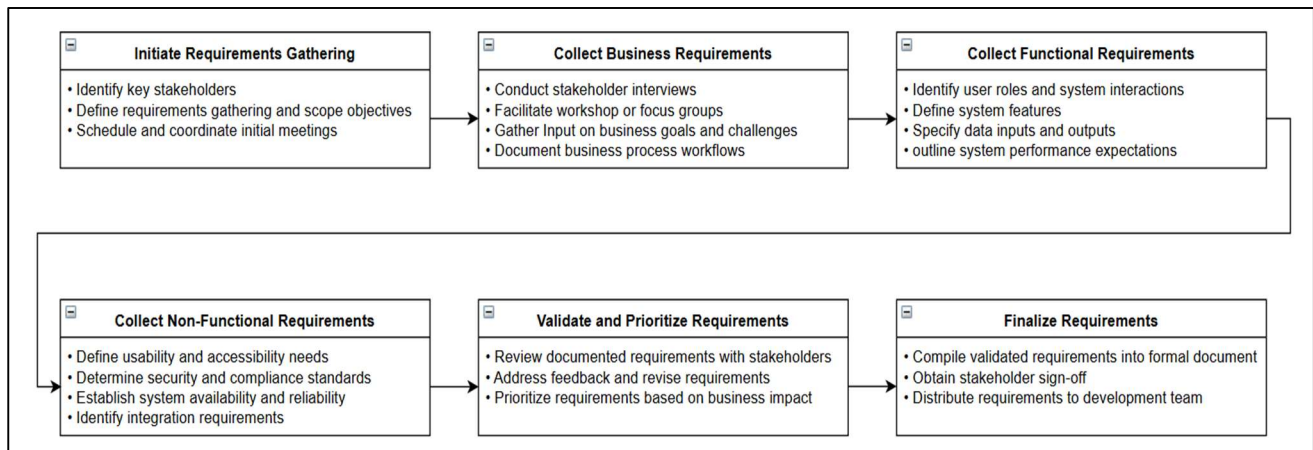
Note. This table outlines the primary stakeholders involved in the HPS customer ordering system development and summarizes their responsibilities in ensuring the project's success.

High-Level WBS for Requirements Gathering

According to Valacich and George (2020), the requirements determination phase is critical to system development, as it established a solid foundation for design and implementation. Effective requirement gathering involves identifying and documenting what users need from a system through a combination of interactive and observational methods.

Figure 1

Work Breakdown Structure (WBS) for Requirements Gathering Phase



Note. This network diagram illustrates the key tasks involved in gathering requirements for the HPS customer ordering system project.

Potential Risks

“Risk management is one of the most challenging disciplines in project management and it's also one of the most important” (Dionisio, 2021, 0:01). There are several potential risks that could impact the success of the customer ordering system development. It is essential to identify and address risks early on in order to create proactive strategies to mitigate potential challenges (Schwenk, 2025). Table 2 outlines the six top risks associated with the development of the customer ordering system, their risk scores, and strategies to mitigate them.

Table 2

Risk Assessment for HPS's Customer Ordering System

Risk	Risk Score	Mitigation Strategy
Integration Challenges with existing HPS infrastructure	4	Conduct detailed compatibility assessments during planning phase
Security vulnerabilities in online transactions and data storage	3	Apply strong encryption standards and follow compliance regulations
Scope creep due to evolving stakeholder requirements	2	Conduct regular scope reviews and require formal sign-off on changes
Insufficient budget to cover development, testing, and training	2	Establish funds and monitor costs throughout the project
Poor usability affecting customer experience and order completion	3	Conduct usability testing with target users and respond based on feedback
Delayed availability of key team members during development	1	Plan flexible schedules and avoid high-risk periods when vacation and workloads are more likely

Note. This table identifies six primary risks associated with the development of the customer ordering system for HPS. Each risk is categorized by type, risk score, and strategies to mitigate them.

By proactively identifying and addressing these key risk areas, HPS can significantly increase the likelihood of a successful customer ordering system development. With thoughtful planning and support, the organization can ensure that the system fulfills technical requirements but also aligns with the shift to a D2C model and long-term business objectives.

Requirements Analysis

System Requirements

To support HPS's transition to a D2C strategy, system requirements must align with the organization's strategic goals and operational needs. Each requirement is designed to address current inefficiencies, support long-term scalability, and enhance the customer experience. This section outlines what the system requires and identifies the primary stakeholders who will rely on it.

A clear approach to defining requirements helps improve understanding and supports choosing the right development solution. HPS should consider using a scoring model to objectively assess system features during development. These requirements represent the primary capabilities ("must haves") needed to ensure successful implementation and long-term value (Stair & Reynolds, 2018). They should perform as the foundation for the development of the customer ordering system.

1. User Experience and Customer Engagement

The system must provide a seamless, computer/mobile-friendly shopping experience. Vogels (2021) found that over a quarter of low-income adults (less than \$30,000 per year) are smartphone-only users. By providing multiple outlets in which the consumer can access the online store, HPS will be able to reach a wider audience. Additionally, customers should be able to browse inventory, add items to a cart, check out as a guest or account holder, and receive timely updates on their orders. Optional account creation will allow users to store order history and shipping details. This will improve convenience while promoting return visits. These features will collectively reduce cart abandonment and support customer

retention (Sani et al., 2024). The primary stakeholders for this requirement will be System Analysts, Marketing & Sales, Customer Service, and IT.

2. Secure Payment Transactions and Accuracy

Secure, flexible payment processing must be built into the checkout process. This includes integrating a PCI DDS that supports various forms of payment (credit cards & PayPal). Research by Baymard Institute (2025) indicates that 10% of cart abandonment was due to customers feeling there were not enough payment methods provided. Furthermore, the system must be thoroughly tested to ensure transaction accuracy, reducing risk and enhancing customer trust. F&A will be responsible for the accuracy of automated tax calculations while IT Support and System Analyst manage technical aspects and troubleshooting.

3. Inventory Integration and Order Fulfillment

Real-time inventory visibility is critical. The system must connect to existing databases to reflect current stock availability across all channels. Automated reconciliation processes should update the inventory database after each sale to ensure consistency between both systems (Team Omniful, 2024). This supports fulfillment accuracy and minimizes order errors. Warehouse and Shipping teams must be notified of the purchase order to initiate order processing and ensure accurate, on-time fulfillment. System Analysts, Operations, and IT Support will be key stakeholders as well.

4. Data Visibility, Reporting, and Access Control

To support informed decision-making and operational oversight, the customer ordering system should provide real-time dashboards and reports on metrics such as sales performance, order trends, and customer behavior. Robust role-based access controls must be implemented to ensure users only access data relevant to their responsibilities (Stair & Reynolds, 2018). Additionally, system logging should be used to monitor activity and support compliance with privacy and security standards. HR, F&A, IT Support, and Cybersecurity will be the stakeholders responsible for access control and compliance while Leadership and Marketing will primarily focus on the insights pulled from dashboards reports.

By organizing requirements into these four high-impact categories, HPS can focus their efforts on features that directly support strategic goals while maintaining clarity around stakeholder ownership. These requirements provide a practical foundation for the system's design and long-term success in the D2C space.

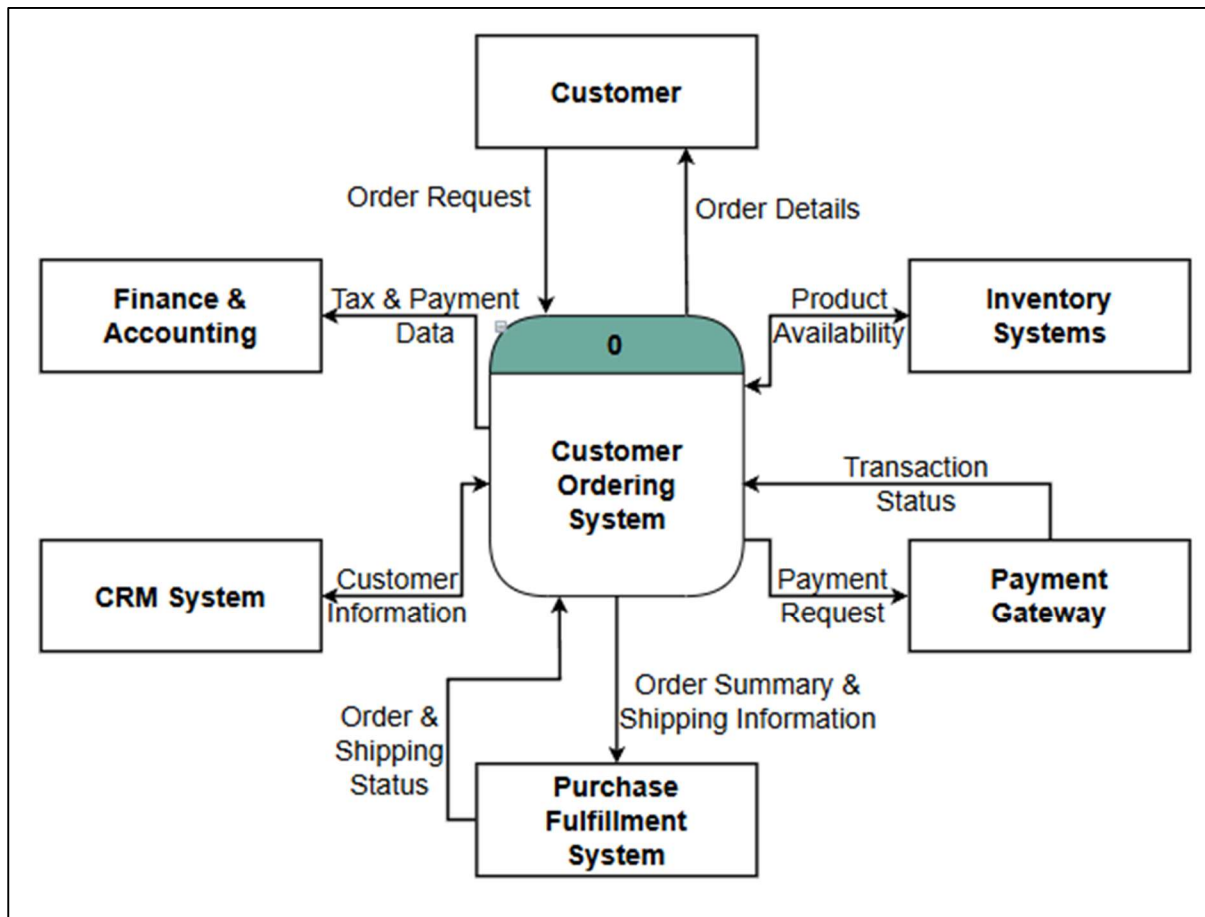
Data Flow Diagram

To better understand and document the flow of information within the HPS customer ordering system, this section includes a series of data flow diagrams (DFDs). These diagrams provide visual representation of how data moves through the system. The process model identifies key processes, data flows, external entities (sources/sinks), and data stores involved in the ordering workflow (Valacich & George, 2020).

The context diagram (level 0) offers the highest-level view of the system, showing how HPS interacts with external entities such as customers, F&A, the customer relations management System (CRM), the purchasing fulfillment system, inventory systems, and the payment gateway (Valacich & George, 2020). It defines the system boundary and the main data exchanges. The context diagram is represented in Figure 2.

Figure 2

Context Diagram for HPS Customer Ordering System



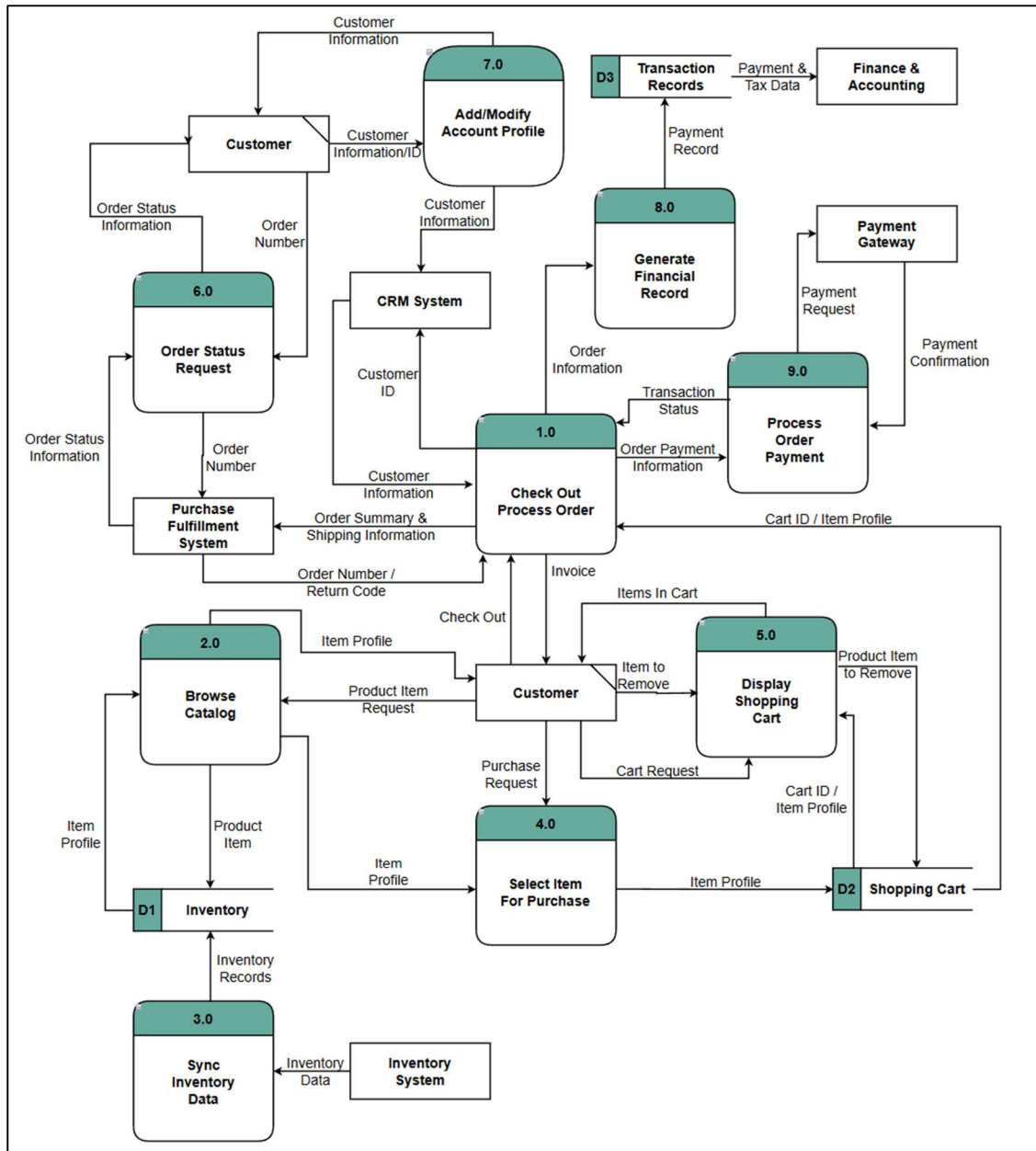
Note. This context diagram illustrates the highest-level of interaction between the HPS customer ordering system and external entities involved with the ordering process.

Figure 3 is the level 1 DFD which expands this by breaking down the main process into its core sub process (Valacich & George, 2020). The diagram reveals how data is transformed and transferred internally. This includes actions such as verifying customer orders, processing payments, updating inventory, and managing customer accounts.

Together, these diagrams help clarify system requirements, promote better stakeholder understanding, and serve as a foundation for future system design and development.

Figure 3

Level 1 DFD for HPS Customer Ordering System



Note. This level 1 data flow diagram decomposes the overall customer ordering process into its major subprocesses and internal data flows.

Economic Feasibility Analysis

HPS must assess the financial benefits and costs associated with the customer purchasing system project. This is commonly referred to as cost-benefit analysis, or economic feasibility. This is a crucial step in the development process as it indicates whether the system is financially viable (Valacich & George, 2020).

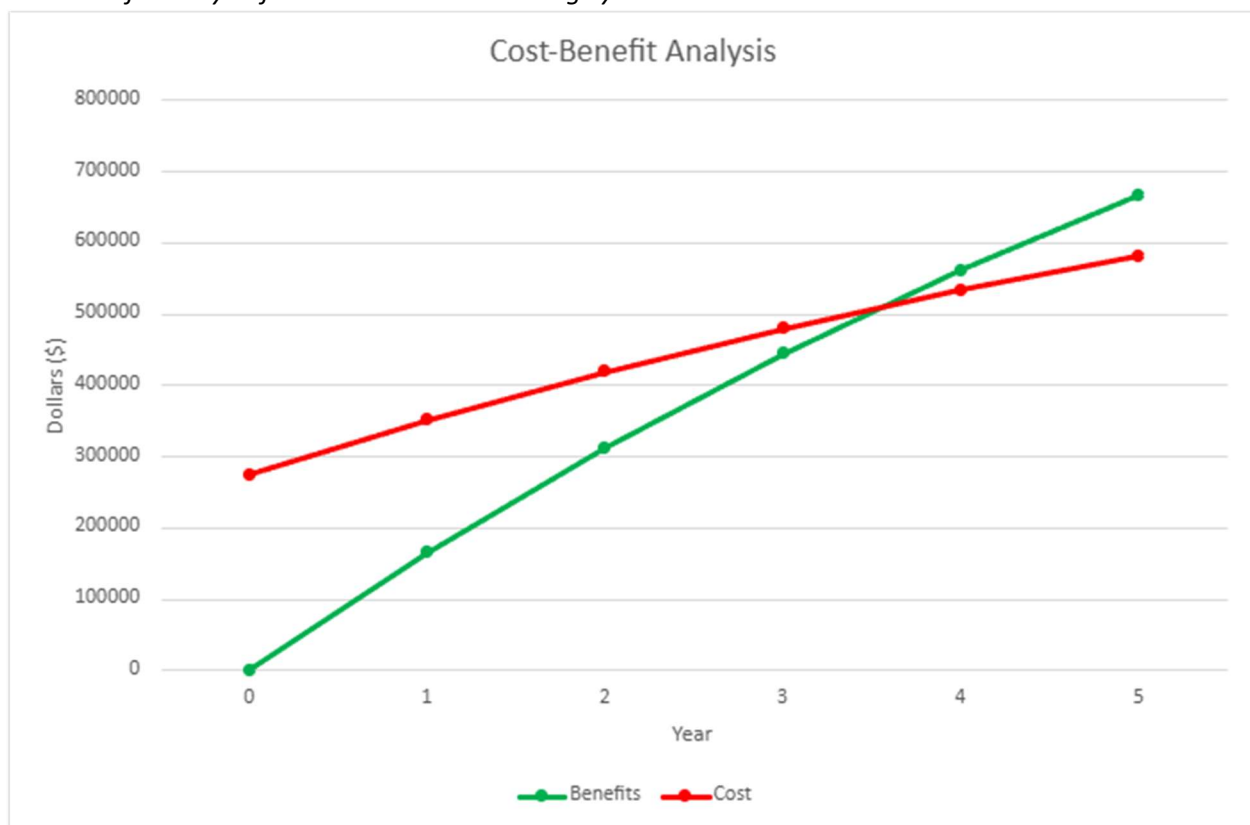
The economic feasibility analysis shows there is a projected Net Present Value (NPV) of \$85,478 and a positive Return on Investment (ROI) of 14.7% over a five-year period. These figures account for an initial

one-time investment of \$275,000, annual benefits of \$180,000, and recurring costs of \$85,000 per year, using a 12% discount rate (see Appendix). These findings suggest that the system's long-term benefits outweigh its costs. Given that the HPS customer ordering system is projected to yield an ROI of approximately 15%, it exceeds the benchmark cited by Penn (2024) and can be considered an excellent return.

The break-even point for the project occurs at approximately 3.5 years, meaning the investment will be fully recovered after the third year of operation. This milestone is critical to evaluate financial risk and support the system's sustainability. Figure 4 illustrates the cumulative benefits and costs over time leading to this break-even point.

Figure 4

Cost-Benefit Analysis for HPS Customer Ordering System



Note. This line graph displays cumulative costs and benefits over a five-year period. The break-even point is reached at approximately 3.5 years.

Development Methodology

To ensure a successful development of HPS's customer ordering system, the organization must adopt a well-suited development methodology. The Scrum framework is an agile approach that is highly effective in managing the project's complexity and evolving requirements. Scrum's iterative sprint cycles allow the HPS development team to deliver functional increments of the customer ordering system quickly while adapting to changing business needs and customer feedback (Valacich & George, 2020).

The clearly defined roles within Scrum will help ensure that stakeholders' priorities are addressed and communication flows smoothly throughout the project lifecycle (Stair & Reynolds, 2018). By promoting transparency, collaboration, and continuous improvement, Scrum aligns well with HPS's goal of creating a flexible, user-friendly customer ordering system that can evolve with the business.

Since Scrum emphasizes individuals and interactions, working software, customer collaboration, and responding to a changing environment, HPS must identify key stakeholders to be a part of the Scrum Team. To help anticipate and remove barriers, keep the project on schedule, and ensure the team delivers results, a Scrum Master must be assigned. This individual will be a skilled Scrum Project Manager. Additionally, an HPS Senior Manager who deeply understands customer needs and business goals will be the Product Owner. This person represents the business side and is responsible for defining the product's vision, prioritizing features, and managing the product backlog. Lastly, the Development Team is responsible for building the product increments during each sprint. For the HPS customer ordering system, this will include anyone who is involved in the design, coding, testing, and delivering the customer ordering system (Stair & Reynolds, 2018).

Adopting the Scrum framework will provide HPS with the structure and flexibility needed to successfully develop its customer ordering system. By clearly defining team roles and embracing an iterative, collaborative approach, HPS can ensure the project remains aligned with business goals, adapts to change, and delivers a high-quality, user-centered solution.

Implementation Plan

Implementation is one of the most costly and time-consuming phases in the systems development life cycle (SDLC) due to the number of people involved and the significant labor required to complete the project (Valacich & George, 2020). Because of this, careful planning is essential to ensure the project stays on track. Implementation should not occur during peak holiday periods when business demand is expected to increase. Therefore, HPS will need to follow a strict schedule, beginning shortly after the Fourth of July (7/4) and completing the implementation before Halloween (10/31).

The following phased implementation plan outlines the strategy for deploying the HPS customer order system, aligning with the goals, stakeholder responsibilities, and agile Scrum methodology discussed earlier. This plan emphasizes flexibility, cross-functional collaboration, and iterative delivery to support HPS's transition to a DTC model.

Phase 1: Strategic Alignment and Sprint Planning (July 7 – July 20)

Establish a unified project and set the stage for an agile development approach that reflects HPS's goals and technical infrastructure.

Tasks:

- Conduct a cross-departmental kickoff meeting to align on goals, sprint structure, and deliverables.
- Finalize roles within the Scrum framework, including Product Owner, Scrum Master, and Development Team.
- Review and confirm project scope, budget, available infrastructure, and timeline consistency with resource assumptions.

- Develop the initial product backlog based on the requirements (e.g. cart functionality, account creation, etc.).

Key Stakeholders:

Scrum Team, Executive Leadership, Finance, IT Support

Phase 2: Feature Development and System Integration (July 21 – August 17)

Translate system requirements into functional testable product increments while ensuring early integration with existing systems (inventory, CRM, payment gateways, etc.).

Tasks:

- Design and build the online ordering interface with mobile compatibility.
- Implement and test PCI DDS-compliant payment gateways (credit card, PayPal).
- Integrate with HPS's inventory database, ensuring live product availability.
- Begin the setup of dashboards and role-based access controls to support internal reporting and compliance needs.

Key Stakeholders:

System Analysts, IT Support, Marketing/Sales, Finance & Accounting

Phase 3: Quality Assurance and Pre-Launch Testing (August 18 – September 14)

Validate system performance against the requirements and confirm readiness for customer use through rigorous testing.

Tasks:

- Conduct unit and integration testing across modules (e.g. payment, inventory, order processing, etc.).
- Simulate real-world scenarios in UAT with involvement from Customer Service to access usability and transaction flow.
- Validate cart completion tracking.
- Verify compliance with security protocols and accuracy of tax/shipping calculations.

Key Stakeholders:

Quality Assurance Team, Customer Service, System Analysts, Finance & Accounting

Phase 4: User Training and Operational Readiness (September 15 – September 30)

Prepare internal teams to support the new system effectively, minimizing disruption during public launch.

Tasks:

- Deliver role-specific training based on stakeholder responsibilities identified in Table 1.
- Provide a training environment where Customer Service and Marketing teams can practice workflows such as order lookups, refunds, and customer communications.
- Distribute guides and job aids for reference.
- Establish a help desk for post launch support.

Key Stakeholders:

System Analysts, IT Support, Customer Service, Warehouse, Learning & Development (L&D)

Phase 5: System Launch and Live Monitoring (October 1 – October 14)

Deploy the ordering system to live customers and closely monitor usage to ensure operational stability.

Tasks:

- Launch the customer ordering system on the HPS website and route live customer traffic to the new system.
- Monitor system logs for errors, dropped transactions, or cart abandonment patterns.
- Support the fulfillment process in coordination with inventory updates.
- Gather immediate customer feedback and internal user reports to identify unforeseen issues.

Key Stakeholders:

System Analysts, IT Support, Customer Service, Warehouse

Phase 6: Post-Launch Evaluation and Optimization (October 15 – October 31)

Evaluate system performance against business objectives and lay the groundwork for continuous improvement.

Tasks:

- Review cart completion rate, system uptime, support requests, and user behavior analytics.
- Conduct review sessions to document lessons learned and identify priorities for future development cycles.
- Update documentation and adjust training materials based on user behavior and feedback.
- Define the next set of features for development, including personalized marketing tools and loyalty system programs.

Key Stakeholders:

Scrum Team, Executive Leadership, Marketing/Sales, Finance & Accounting, Quality Assurance.

Training Plan

A Training Plan is necessary to ensure that all relevant HPS staff can effectively and efficiently use the new customer order system. The training will support internal users across departments in navigating the system, assisting customers, and maintaining operational continuity.

Audience and Training Scope

Table 3

Role-Based Training Overview for HPS Customer Ordering System

Audience	Specialized Training	Key Topics Covered
Customer Service	Manage orders, resolve issues, and supporting customer accounts	Order lookups, refunds and returns, account creation, order status updates, FAQs
Marketing & Sales	Utilize dashboards and customer data for campaigns and insights	Using reports, analyzing customer behavior, applying promo codes, tracking conversations
Finance & Accounting	Ensuring payment/tax accuracy and compliance	Reviewing tax/shipping calculations, transaction logs, financial reporting access
IT Support	Troubleshooting system issues, managing roles, and access control	User permissions, system monitoring, help desk
Warehouse & Shipping	Fulfilling orders and confirming inventory availability	Order notification, inventory sync, fulfillment accuracy, shipping updates

Note. This table outlines the primary internal user groups, their specialized training focus, and the system features and functions most relevant to their core responsibilities.

Training Method

Table 4

Training Methods and Resources Required for HPS Customer Ordering System

Training Method	Description	Resources Required
Live Workshops	Instructor led walkthroughs conducted in-person and/or virtually	Conference room or video access, projector or screen sharing tools, attendance tracking
Training Systems	Simulated version of the system that allows users to practice real tasks	Access to training environment, demo accounts, sample data sets
Video Modules	Task-based tutorials for reference and self-paced learning	Learning Management System (LMS) or shared drive, video hosting platform, audio device
Reference Guides	Printable standard operating procedures (SOP)	PDFs and/or printable handouts, SharePoint for ongoing updates

Note. This table outlines the delivery methods and corresponding resources required for training stakeholders on the HPS Customer Ordering System.

Training Schedule

Table 5

Training Schedule and Delivery Plan for HPS Customer Ordering System

Date	Session Title	Key Stakeholder	Format	Trainers
Sept 15	System Overview & Navigation	All Users	Live Workshop (2 hrs)	Scrum Master, Product Owner, System Analyst
Sept 16-18	Specialized Training	By Function	Live Workshop (2 hr per dept), Training Systems, Video Modules, Reference Guide	System Analysts
Sept 19	Error Handling & Support Flow	All Users	Video Modules, Reference Guides	Scrum Master, IT Support, Customer Service Lead
Sept 22	Cross-Functional Workflow Simulation	All Users	Live Workshop (1.5 hrs), Training Systems	Scrum Master, Product Owner, IT Support
Sept 23	Live Q&A and Feedback Session	All Users	Live Workshop (1 hour)	Scrum Master, System Analysts
Sept 29-30	Optional Targeted Follow-Up Training	By Request	Live Workshop, Training Systems	Scrum Master, applicable subject matter experts

Note. This schedule outlines the key training sessions, participants, delivery formats, and facilitators to support user adoption and successful system implementation at HPS.

Evaluation

- Quizzes for each training module
- Completion of training system tasks
- Post-training feedback survey
- Live Q&A and feedback session

Follow-Up Support

- Designated office hours for one-on-one help
- Targeted follow-up training sessions in response to September 23 feedback.
- IM/Email/Call support monitored by IT and System Analysts
- Refresher videos available on SharePoint

Testing Plan

The purpose of this testing plan is to verify that the HPS customer ordering system meets all functional, security, and performance requirements. Testing is an essential part of the SDLC, providing the opportunity to validate that the system performs as intended before deployment while reducing the risk of defects in production (Valacich & George, 2020). Consistent with Scrum, testing will occur iteratively throughout the development sprints to detect issues early, validate backlog items, and ensure high system quality at launch (Stair & Reynolds, 2018).

Objectives

- Validate that all system features meet the business requirements and acceptance criteria outlined in the product backlog.
- Confirm secure and accurate payment processing.
- Ensure real-time inventory sync functions.
- Verify usability, accessibility, and mobile responsiveness to support a seamless user experience.
- Conduct UAT with functional stakeholders.
- Support ongoing optimization based on continuous testing feedback.

Scope

In Scope:

- Functional testing of all customer workflows: browsing, cart management, checkout, account creation, and order confirmation
 - Integration testing with external systems including:
 - Payment gateways (credit cards, PayPal)
 - Inventory management system
 - CRM system
- Tax and shipping calculation validation
- Mobile and desktop compatibility on all major browsers (Chrome, Safari, Edge, etc.)
- Role-based access control testing
- UAT involving Customer Service, Marketing, and F&A
- Dashboard data validation

Out of Scope:

- Stress testing for high concurrency beyond expected load
- Third-party plugins or apps not included in the approved backlog
- Testing Approach & Timeline

Testing Strategy & Timeline

Table 6

Testing Approach and Timeline for HPS Customer Ordering System

Testing Type	Description	Planned Timeline
Sprint Testing	Product backlog validation conducted at the end of each sprint	July 21 – Sept 5 (ongoing)
Unit Testing	Developer led testing of individual functions and components	July 21 – Sept 5
Integration Testing	Validate data flow between systems	Aug 11 – Aug 29
System Testing	End-to-end testing of workflows, reporting, and order lifecycle	Aug 25 – Aug 29
User Acceptance Testing	Testing of customer order flows	Sept 1 – Sept 5
Regression Testing	Confirm updates do not disrupt existing functionality	Sept 8 – Sept 11
Final Review	Final walkthrough, metric validation, and formal approval	Sept 12 – Sept 14

Note. This table outlines the timing and intent for each test to ensure system quality and readiness ahead of the October 1 launch. The Final Review concludes Phase 3 and marks the formal system sign-off before training begins.

Test Environment

- **Test Systems:** Mirrors production setup with test versions of all integrated systems.
- **Test Account/Data:** Includes sample users, inventory items, and mock payment methods.
- **Device/Browser:** Desktop and mobile devices

Sample Test Case

Table 7

Sample Test Cases for HPS Customer Ordering System

Test ID	Description	Expected Results
TC001	Add items to cart as guest	Items appear in cart, total calculates accurately
TC002	Checkout with PayPal	Redirect succeeds, confirmation received, order recorded
TC003	Register account during checkout	User receives account confirmation and sees order in history
TC004	Out-of-stock item attempted to purchase	System prevents order, shows real-time inventory updates
TC005	Tax and shipping calculated	Accurate tax/shipping
TC006	Admin generates sales report	Dashboard loads data and exports correctly

Note. This table outlines test scenarios and expected outcomes used to validate functionality during the implementation phase.

Testing Roles & Resources

Table 8

Testing Roles and Resources for the HPS Customer Ordering System

Role	Responsibilities	Resources Required
Scrum Master	Facilitates spring testing, tracks issue resolution, ensure product backlogs are completed	1 Scrum Master
Product Owner	Defines acceptance criteria, approves completed product backlog, reviews test outcomes during sprint	1 Product Owner
System Analysts	Conduct integrations testing across all systems, support UAT environment setup	2 System Analysts
Developers	Fix bugs, support unit testing, assist with troubleshooting	2 Developers
QA Team	Coordinates/Executes all test phases, maintain documentation, support system and regression testing	1 QA Lead 2 QA Analysts
Business Stakeholders	Participate in IAT, provide usability and workflow feedback	5 Department Heads (CS, Marketing/Sales, F&A, IT, Warehouse)
IT Support	Maintain test environment, monitor performance, address technical issues	1 IT Support Specialist

Note. This table identifies the key roles and required personnel involved in testing.

Risk & Mitigation

Table 9

Risk & Mitigation Strategies for HPS Customer Ordering System

Risk	Mitigation Strategy
Delayed Bug Resolution	Track issues in sprint, increase developer testing
Unclear Acceptance Criteria	Review backlog jointly before sprints
Insufficient UAT Engagement	Pre-schedule UAT time and use testing environment
Integration data mismatch or delay	Perform dry runs and automated sync testing

Note. This table outlines key testing risks and provides proactive mitigation strategies. These approaches help reduce testing bottlenecks and support a stable system launch.

Approval

- Product backlog must pass acceptance criteria and be accepted by the Product Owner.
- All critical and high-priority defects must be resolved prior to launch.
- Final system sign-off provided by the Product Owner, Scrum Master, and QA Lead.

Conclusions

The development of the HPS customer ordering system is a critical step in supporting the organization's transition to a D2C business model. This system will enable the company to modernize its operations, enhance customer engagement, and gain control over the consumer experience. By focusing on

seamless usability, secure transactions, real-time inventory tracking, and personalized communication, the system directly supports HPS's strategic goals and objectives.

The implementation plan outlined is designed to minimize disruption while maximizing cross-functional collaboration, which is supported by the Scrum framework. This approach enables HPS to respond quickly to feedback, adapt to emerging requirements, and deliver functional increments in short, controlled cycles (Stair & Reynolds, 2018). Comprehensive stakeholder involvement, a detailed training strategy, and iterative testing throughout each sprint ensure that the system will meet technical requirements and be adopted successfully by end users.

Looking forward, the flexibility of the system lays the groundwork for future adaptive and perfective maintenance (Valacich & George, 2020). Features such as loyalty programs and targeted marketing can be added in future sprints, ensuring the platform evolves alongside consumer expectations and market demand. Future sprints may introduce AI-driven product recommendations, integration with external marketplaces, or expanded analytics dashboards tailored for marketing automation. By investing in this agile solution, HPS can position itself to compete with competitors in the D2C space.

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Appendix

Economic Feasibility Analysis for HPS Customer Ordering System

HULABALOO PARTY SUPPLIES						
Cost of Capital		12%				
Initial Capital Expense		275,000				
Years		1	2	3	4	5
Cash Inflows (Sales)		185,000	185,000	185,000	185,000	185,000
Present Value (PV) of Cash Inflows		165,179	147,481	131,679	117,571	104,974
Benefits		165,179	312,659	444,339	561,910	666,884
Initial Investment	275,000					
Cash Outflows (Expenses)		85,000	85,000	85,000	85,000	85,000
Present Value (PV) of Cash Outflows	275,000	75,893	67,761	60,501	54,019	48,231
Costs	275,000	350,893	418,654	479,156	533,175	581,406
NPV (Net Present Value)	85,478					
ROI	14.70188192					
<i>Breakeven Analysis</i>						
Yearly NPV Cash Flow	(275,000)	89,286	79,719	71,178	63,552	56,743
Overall NPV Cash Flow	(275,000)	(185,714)	(105,995)	(34,817)	28,735	85,478
Break-even Ratio	0.54785024					
Break-even Years	3.54					

Note. This appendix presents a five-year discounted cash flow analysis used to assess the economic feasibility of the HPS customer ordering system.